



Syllabus Reference:

AHL 5.18

Content	Guidance, clarification and syllabus links
First order differential equations. Numerical solution of $\frac{dy}{dx} = f(x, y)$ using Euler's method.	$x_{n+1} = x_n + h$, where h is a constant.
Variables separable.	<i>Example:</i> the logistic equation $\frac{dn}{dt} = kn(a - n)$, $a, k \in \mathbb{R}$ Link to: partial fractions (AHL1.11) and use of partial fractions to rearrange the integrand (AHL5.15).
Homogeneous differential equation $\frac{dy}{dx} = f\left(\frac{y}{x}\right)$ using the substitution $y = vx$.	
Content	Guidance, clarification and syllabus links
Solution of $y' + P(x)y = Q(x)$, using the integrating factor.	

Formula

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Euler's method

$y_{n+1} = y_n + h \times f(x_n, y_n)$; $x_{n+1} = x_n + h$, where h is a constant (step length)

Question(1)–Paper–2



[Maximum mark: 5]

Consider the differential equation $\frac{dy}{dx} - e^x = 1$.

Given that $y(0) = 1$, use Euler's method with step length $h = 0.25$ to find an approximation for $y(1)$. Give your answer correct to two decimal places.

$$\frac{dy}{dx} = 1 + e^x$$

$$y(0) = 1, \quad h = 0.25 \quad (\text{Step size})$$

Formula: $y_{n+1} = y_n + h \times f(x_n, y_n), \quad x_{n+1} = x_n + h$

Where: $f(x_n, y_n) = \frac{dy}{dx} = 1 + e^x$

To find: $y(1)$

Substituting the given values

$$y(0) = 1 \quad (\text{initial values})$$

$$x_{n+1} = x_n + 0.25$$

$$y_{n+1} = y_n + 0.25 \times \frac{dy}{dx}$$

In GDC, we label column A as "x"
column B as "y"
and column C as "f"

$\frac{dy}{dx}$

	A x	B y	C f	
R1	0	1	=	}
R2	0.25			
R3				
R4				
R5				

$$= 1 + e^{a_1}$$



In GDC, first row, column A type (0)
first row, column B type (1)

and first row, column C, type $\longrightarrow = 1 + e^{a_1}$

In GDC, second row, we enter (0.25) in column A

Then select row 1 column A, press shift and move cursor to row 2 (so that column A both cells are selected), now...
press menu, data, fill...and drag down arrow key until you get 1 in last row of column A

Now take cursor to column B row 2 & Type $\longrightarrow = b_1 + 0.25c_1$

Then keeping cursor there press menu, data, fill...and drag down arrow key until last row

Note that you won't see any values in column B

Also note that in row 1 $\longrightarrow a_1 = x_0$, $b_1 = y_0$ & $c_1 = 1 + e^{x_0}$



Now, take cursor to **first** row of column C, and press, **menu, data, fill...** and drag cursor down to last row

Now all values will be filled, the **answer** is in column B, last row (where column A has value **(1)**)

You have to draw following table on exam paper

	A x	B y	C f	D
1	0	1	2	
2	0.25	1.5	2.28403	
3	0.5	2.07101	2.64872	
4	0.75	2.73319	3.117	
5	1	3.51244	3.71828	

B2 = $b1 + 0.25 \cdot c1$

	A x	B y	C f	D
1	0	1	2	
2	0.25	1.5	2.28403	
3	0.5	2.07101	2.64872	
4	0.75	2.73319	3.117	
5	1	3.51244	3.71828	

C1 = $1 + e^{a1}$

$$\therefore y(1) \approx 3.51$$

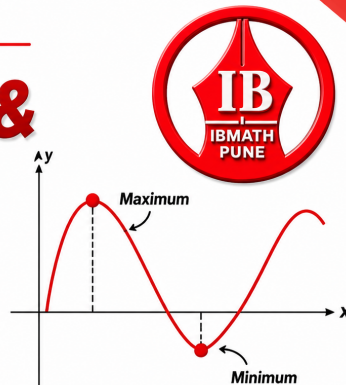


Practice Questions, PDF link is given in the description of this video.



How to find **MAXIMUM & MINIMUM VALUES**

4 EASY STEPS



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